

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

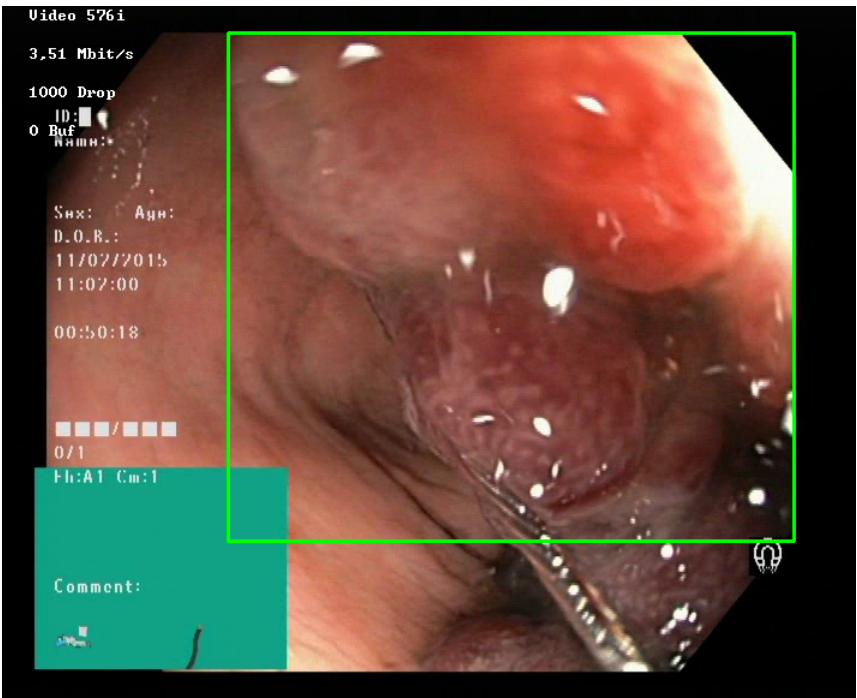
Video ID:	3b0fab03-6367-4bff-9272-7ed2ba0dfb31
Total Frames:	4744
Frame Rate:	25.0 fps (assumed)
Duration:	189.8 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	2444	✓
RT-DETR Detections	3997	✓
MedSAM2 Detections	4052	✓
Consensus (3-Model Agreement)	2	✓
Risk Classification	Count	Percentage
HIGH RISK	1	50.0%
MEDIUM RISK	1	50.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	2	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 95 frames)



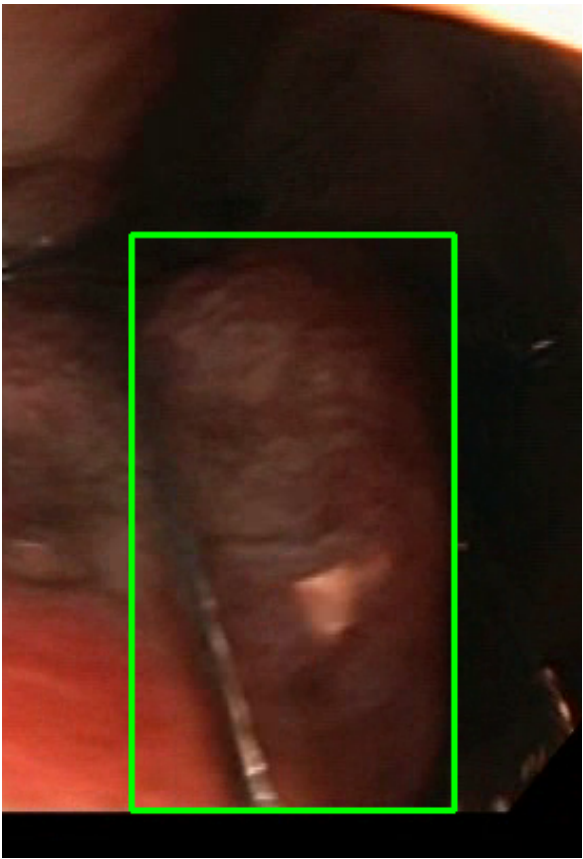
Feature	Value	Interpretation
Frame Index	2370	Frame 2370 of 4744
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.231	Color-based vascularization
Relative Radius	26.0%	Polyp size as % of frame diagonal
Texture Score	0.091	Surface morphology
Vessel Visibility	0.907	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	ADENOMATOUS POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	83.0%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 1646 frames)



Feature	Value	Interpretation
Frame Index	355	Frame 355 of 4744
Detection Method	Detector Consensus (YOLO / RT-DETR / MedSAM2)	Detector Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 81.9%
Redness Score	0.102	Color-based vascularization

Relative Radius	42.7%	Polyp size as % of frame diagonal
Texture Score	0.255	Surface morphology
Vessel Visibility	0.064	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	86.4%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 2 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (50.0%)
- MEDIUM RISK polyps: 1 (50.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.